Deduction theorem for the axiomatic calculus

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs ax:lif1, ax:lif2, prop:mp, prop:monotonicity, prop:reflexivity, ex:identity; upstream text artifact '!!^a' at source line 664 (a mangled capital A before 'derivation of B') is the author's own text delivered verbatim - flagged to the correction layer.

Editorial note. The label thm:deduction-thm names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Axiom instances and modus ponens. Same unit, ax:lif1 / ax:lif2 (propositional axiom group), prop:mp. Basis and inductive step: Hilbert schema instances plus two applications of MP.
- Structural facts of the derivability relation. Same unit, prop:monotonicity / prop:reflexivity / ex:identity. Weakening for the easy direction; the identity derivation for the A -> A basis case.

Source reference. Original proposition prop-mp: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 625–625. Prerequisite pointer invoked at proof line(s) 651; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ex-identity: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 1519–1519. Prerequisite pointer invoked at proof line(s) 666; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lif1: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 486–486. Prerequisite pointer

invoked at proof line(s) 662; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lif2: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 487–487. Prerequisite pointer invoked at proof line(s) 686; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-monotonicity: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 832–832. Prerequisite pointer invoked at proof line(s) 650; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-reflexivity: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 823–823. Prerequisite pointer invoked at proof line(s) 651; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Deduction Theorem)**.** $\Gamma \cup \{A\} \vdash B$ *if and only if* $\Gamma \vdash A \to B$.

Proof. The "if" direction is immediate. If $\Gamma \vdash A \to B$ then also $\Gamma \cup \{A\} \vdash A \to B$ by prop-monotonicity. Also, $\Gamma \cup \{A\} \vdash A$ by prop-reflexivity. So, by prop-mp, $\Gamma \cup \{A\} \vdash B$.

For the "only if" direction, we proceed by induction on the length of the derivation of B from $\Gamma \cup \{A\}$.

For the induction basis, we prove the claim for every derivation of length 1. derivation of B from $\Gamma \cup \{A\}$ of length 1 consists of B by itself; and if it is correct B is either $\in \Gamma \cup \{A\}$ or is an axiom. If $B \in \Gamma$ or is an axiom, then $\Gamma \vdash B$. We also have that $\Gamma \vdash B \to (A \to B)$ by ax-lif1, and prop-mp gives $\Gamma \vdash A \to B$. If $B \in \{A\}$ then $\Gamma \vdash A \to B$ because the last sentence $A \to B$ is the same as $A \to A$, and we have derived that in ex-identity.

For the inductive step, suppose derivation of B from $\Gamma \cup \{A\}$ ends with a step B which is justified by modus ponens. (If it is not justified by modus ponens, $B \in \Gamma$, $B \equiv A$, or B is an axiom, and the same reasoning as in the induction basis applies.) Then some previous steps in the derivation are $C \to B$ and C, for some formula C, i.e., $\Gamma \cup \{A\} \vdash C \to B$ and $\Gamma \cup \{A\} \vdash C$, and the respective derivations are shorter, so the inductive hypothesis applies to them. We thus have both:

$$\Gamma \vdash A \to (C \to B);$$
$$\Gamma \vdash A \to C.$$

But also

$$\Gamma \vdash (A \to (C \to B)) \to ((A \to C) \to (A \to B)),$$

by ax-lif2, and two applications of prop-mp give $\Gamma \vdash A \to B$, as required. □